

Steam Database Analysis

All of the original functions were copied and pasted from this website:

<https://nik-davis.github.io/posts/2019/steam-data-collection/>

This page has useful data for making graphs:

<https://nik-davis.github.io/posts/2019/steam-data-exploration/>

In [4]:

```
import csv
from matplotlib import pyplot as plt
import numpy as np
import pandas as pd
```

In [140...]

```
import numpy as np
from matplotlib import pyplot as plt
import pandas as pd

def extract_players_vs_time(data):
    data = data.replace("M", "")
    data = data.replace("L", "")
    data = data.split(' ')
    data = list(filter(lambda a: a != "", data)) # remove all instances of " " from the
    data = np.array(data).astype(float)
    x = data[0:-2:2] # SyntaxError: invalid syntax
    y = data[1:-1:2]
    print(f'{len(x) = }')
    return x, y

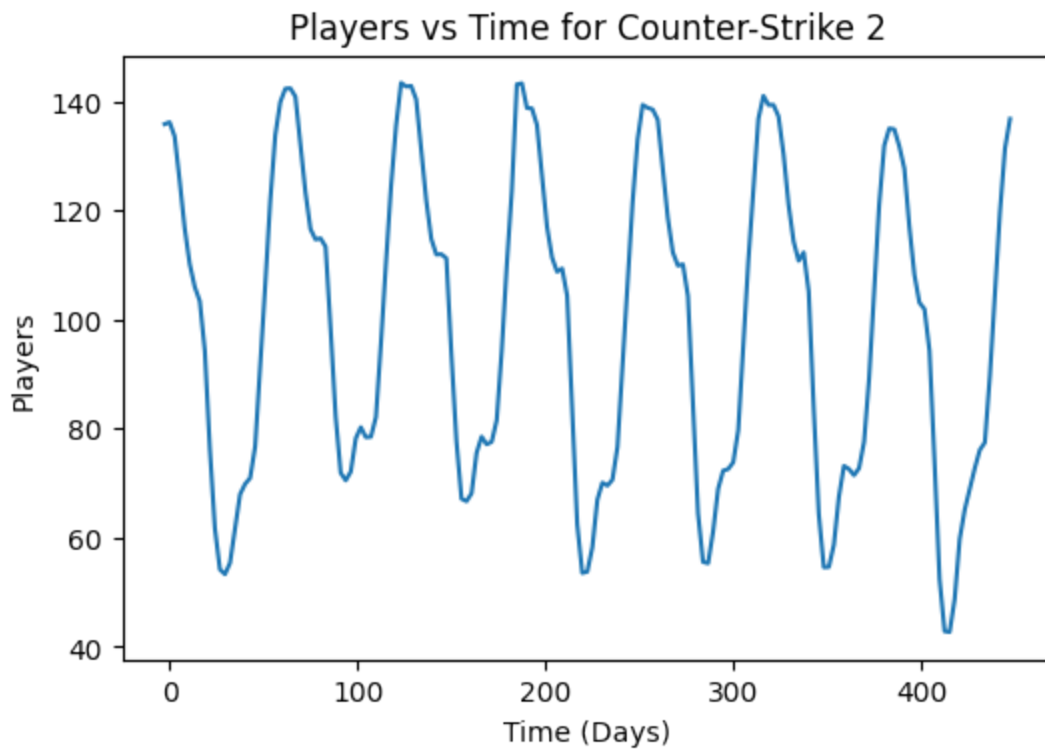
def plot_players_vs_time(x, y, name):
    plt.figure(dpi=100)
    plt.title(f"Players vs Time for {name}")
    plt.plot(x, y)
    plt.xlabel('Time (Days)')
    plt.ylabel('Players')
    plt.show()

games = pd.DataFrame()
```

In [141...]

```
name = "Counter-Strike 2"
data = "M -2.6785714285714 135.849186 L 0 136.204893 L 2.6785714285714 133.618815 L 5.3
x, y = extract_players_vs_time(data)
plt.show()
games = games.append({"Name": name, "Players Vs Time": y}, ignore_index=True)
plot_players_vs_time(x, y, name)
```

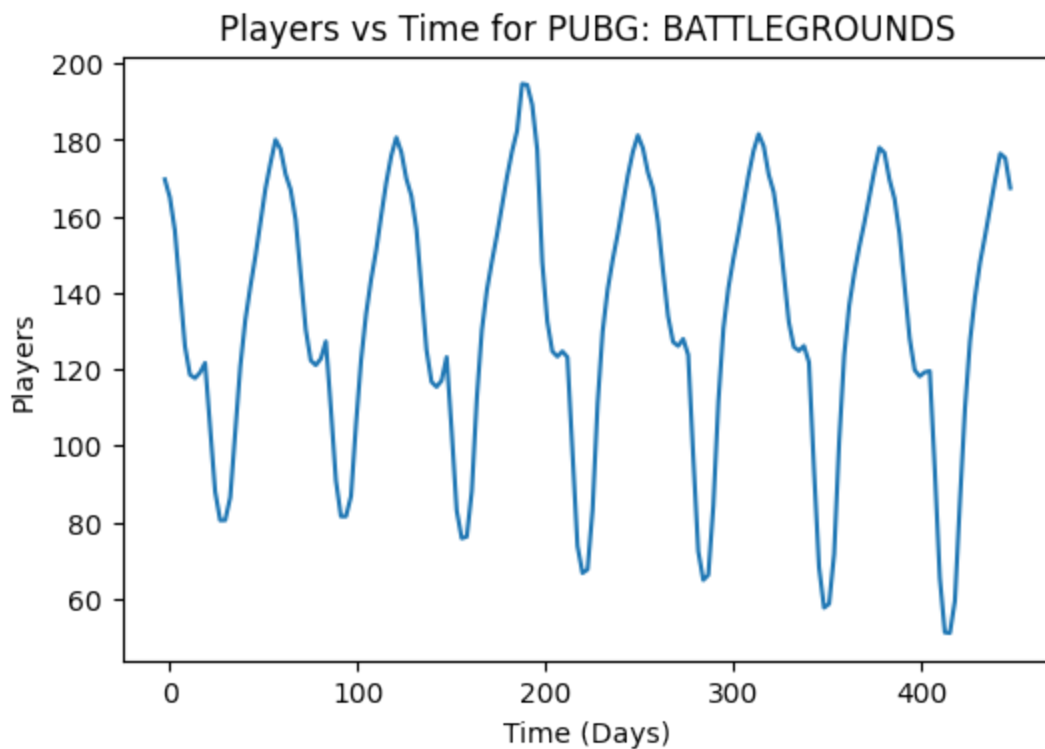
len(x) = 169



In [142...

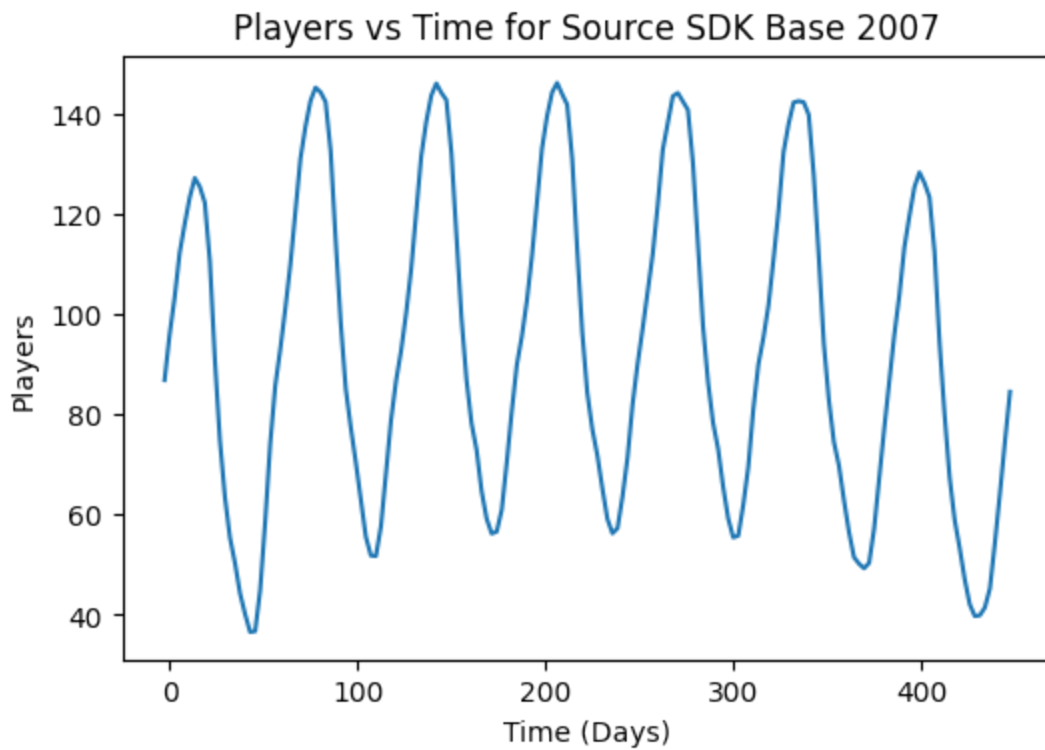
```
name = "PUBG: BATTLEGROUNDS"
data = "M -2.6785714285714 169.639866 L 0 164.96865 L 2.6785714285714 156.292884 L 5.35"
x, y = extract_players_vs_time(data)
plt.show()
games = games.append({"Name": name, "Players Vs Time": y}, ignore_index=True)
plot_players_vs_time(x, y, name)
```

len(x) = 169



In [143...

```
name = "Dota 2"
data = "M -2.6785714285714 115.025064 L 0 111.54 L 2.6785714285714 108.129384 L 5.357142857142857 105.25064 L 8.035714285714286 102.37142857142857 L 10.714285714285714 99.49285714285714 L 13.392857142857143 96.61428571428571 L 16.07142857142857 93.73571428571429 L 18.75 90.85714285714286 L 21.42857142857143 87.97857142857143 L 24.107142857142858 85.1 L 26.785714285714285 82.22142857142857 L 29.464285714285715 79.34285714285714 L 32.14285714285714 76.46428571428571 L 34.82142857142857 73.58571428571429 L 37.5 70.70714285714286 L 40.17857142857143 67.82857142857143 L 42.857142857142854 64.95 45.535714285714286 62.07142857142857 L 48.21428571428571 59.19285714285714 L 50.89285714285714 56.31428571428571 L 53.57142857142857 53.43571428571428 L 56.25 50.55714285714285 L 58.92857142857143 47.67857142857143 L 61.607142857142854 44.8 L 64.285714285714286 41.92142857142857 L 66.96428571428571 39.04285714285714 L 69.64285714285714 36.16428571428571 L 72.32142857142857 33.285714285714286 L 75 30.407142857142858 L 77.67857142857143 27.52857142857143 L 80.35714285714286 24.649285714285714 L 83.03571428571429 21.77142857142857 L 85.71428571428571 18.892857142857142 L 88.39285714285714 16.014285714285714 L 91.07142857142857 13.135714285714286 L 93.75 10.257142857142858 L 96.42857142857143 7.378571428571429 L 99.10714285714286 4.5 101.78571428571429 1.6214285714285714 L 104.46428571428571 -1.2571428571428571 L 107.14285714285714 -4.135714285714286 L 109.82142857142857 -7.014285714285714 L 112.5 9.857142857142857 L 115.17857142857143 -1.9214285714285714 L 117.85714285714286 -4.8 L 120.53571428571429 -7.678571428571429 L 123.21428571428571 -10.557142857142858 L 125.89285714285714 -13.435714285714286 L 128.57142857142857 -16.314285714285714 L 131.25 -19.192857142857142 L 133.92857142857143 -22.07142857142857 L 136.60714285714286 -24.95 L 139.28571428571429 -27.82857142857143 L 141.96428571428571 -30.707142857142858 L 144.64285714285714 -33.585714285714286 L 147.32142857142857 -36.46428571428571 L 150 39.34285714285714 L 152.67857142857143 42.22142857142857 L 155.35714285714286 45.1 L 158.03571428571429 47.97857142857143 L 160.71428571428571 50.857142857142858 L 163.39285714285714 53.735714285714286 L 166.07142857142857 56.61428571428571 L 168.75 59.49285714285714 L 171.42857142857143 62.37142857142857 L 174.10714285714286 65.25 L 176.78571428571429 68.129384 L 179.46428571428571 71.00785714285714 L 182.14285714285714 73.88571428571429 L 184.82142857142857 76.76428571428571 L 187.5 79.64285714285714 L 190.17857142857143 82.52142857142857 L 192.85714285714286 85.4 L 195.53571428571429 88.27857142857143 L 198.21428571428571 91.15714285714286 L 200.89285714285714 94.03571428571429 L 203.57142857142857 96.91428571428571 L 206.25 99.79285714285714 L 208.92857142857143 102.67142857142857 L 211.60714285714286 105.55 L 214.28571428571429 108.42857142857143 L 216.96428571428571 111.30714285714286 L 219.64285714285714 114.18571428571429 L 222.32142857142857 117.06428571428571 L 225 119.94285714285714 L 227.67857142857143 122.82142857142857 L 230.35714285714286 125.7 L 233.03571428571429 128.57857142857143 L 235.71428571428571 131.45714285714286 L 238.39285714285714 134.33571428571429 L 241.07142857142857 137.21428571428571 L 243.75 140.09285714285714 L 246.42857142857143 142.97142857142857 L 249.10714285714286 145.85 L 251.78571428571429 148.72857142857143 L 254.46428571428571 151.60714285714286 L 257.14285714285714 154.48571428571429 L 259.82142857142857 157.36428571428571 L 262.5 160.24285714285714 L 265.17857142857143 163.12142857142857 L 267.85714285714286 166.0 L 270.53571428571429 168.87857142857143 L 273.21428571428571 171.75714285714286 L 275.89285714285714 174.63571428571429 L 278.57142857142857 177.51428571428571 L 281.25 180.39285714285714 L 283.92857142857143 183.27142857142857 L 286.60714285714286 186.15 L 289.28571428571429 189.02857142857143 L 291.96428571428571 191.90714285714286 L 294.64285714285714 194.78571428571429 L 297.32142857142857 197.66428571428571 L 300 200.54285714285714 L 302.67857142857143 203.42142857142857 L 305.35714285714286 206.3 L 308.03571428571429 209.17857142857143 L 310.71428571428571 212.05714285714286 L 313.39285714285714 214.93571428571429 L 316.07142857142857 217.81428571428571 L 318.75 220.69285714285714 L 321.42857142857143 223.57142857142857 L 324.10714285714286 226.45 L 326.78571428571429 229.32857142857143 L 329.46428571428571 232.20714285714286 L 332.14285714285714 235.08571428571429 L 334.82142857142857 237.96428571428571 L 337.5 240.84285714285714 L 340.17857142857143 243.72142857142857 L 342.85714285714286 246.6 L 345.53571428571429 249.47857142857143 L 348.21428571428571 252.35714285714286 L 350.89285714285714 255.23571428571429 L 353.57142857142857 258.11428571428571 L 356.25 261.0 L 358.92857142857143 263.87857142857143 L 361.60714285714286 266.75714285714286 L 364.28571428571429 269.63571428571429 L 366.96428571428571 272.51428571428571 L 369.64285714285714 275.39285714285714 L 372.32142857142857 278.27142857142857 L 375 281.15 L 377.67857142857143 284.02857142857143 L 380.35714285714286 286.90714285714286 L 383.03571428571429 289.78571428571429 L 385.71428571428571 292.66428571428571 L 388.39285714285714 295.54285714285714 L 391.07142857142857 298.42142857142857 L 393.75 301.3 L 396.42857142857143 304.17857142857143 L 399.10714285714286 307.05714285714286 L 401.78571428571429 309.93571428571429 L 404.46428571428571 312.81428571428571 L 407.14285714285714 315.69285714285714 L 409.82142857142857 318.57142857142857 L 412.5 321.45 L 415.17857142857143 324.32857142857143 L 417.85714285714286 327.20714285714286 L 420.53571428571429 330.08571428571429 L 423.21428571428571 332.96428571428571 L 425.89285714285714 335.84285714285714 L 428.57142857142857 338.72142857142857 L 431.25 341.6 L 433.92857142857143 344.47857142857143 L 436.60714285714286 347.35714285714286 L 439.28571428571429 350.23571428571429 L 441.96428571428571 353.11428571428571 L 444.64285714285714 356.0 L 447.32142857142857 358.87857142857143 L 450 361.75714285714286 L 452.67857142857143 364.63571428571429 L 455.35714285714286 367.51428571428571 L 458.03571428571429 370.39285714285714 L 460.71428571428571 373.27142857142857 L 463.39285714285714 376.15 L 466.07142857142857 379.02857142857143 L 468.75 381.90714285714286 L 471.42857142857143 384.78571428571429 L 474.10714285714286 387.66428571428571 L 476.78571428571429 390.54285714285714 L 479.46428571428571 393.42142857142857 L 482.14285714285714 396.3 L 484.82142857142857 399.17857142857143 L 487.5 402.05714285714286 L 490.17857142857143 404.93571428571429 L 492.85714285714286 407.81428571428571 L 495.53571428571429 410.69285714285714 L 498.21428571428571 413.57142857142857 L 500.89285714285714 416.45 L 503.57142857142857 419.32857142857143 L 506.25 422.20714285714286 L 508.92857142857143 425.08571428571429 L 511.60714285714286 427.96428571428571 L 514.28571428571429 430.84285714285714 L 516.96428571428571 433.72142857142857 L 519.64285714285714 436.6 L 522.32142857142857 439.47857142857143 L 525 442.35714285714286 L 527.67857142857143 445.23571428571429 L 530.35714285714286 448.11428571428571 L 533.03571428571429 451.0 L 535.71428571428571 453.87857142857143 L 538.39285714285714 456.75714285714286 L 541.07142857142857 459.63571428571429 L 543.75 462.51428571428571 L 546.42857142857143 465.39285714285714 L 549.10714285714286 468.27142857142857 L 551.78571428571429 471.15 L 554.46428571428571 474.02857142857143 L 557.14285714285714 476.90714285714286 L 559.82142857142857 479.78571428571429 L 562.5 482.66428571428571 L 565.17857142857143 485.54285714285714 L 567.85714285714286 488.42142857142857 L 570.53571428571429 491.3 L 573.21428571428571 494.17857142857143 L 575.89285714285714 497.05714285714286 L 578.57142857142857 499.93571428571429 L 581.25 502.81428571428571 L 583.92857142857143 505.69285714285714 L 586.60714285714286 508.57142857142857 L 589.28571428571429 511.45 L 591.96428571428571 514.32857142857143 L 594.64285714285714 517.20714285714286 L 597.32142857142857 520.08571428571429 L 600 522.96428571428571 L 602.67857142857143 525.84285714285714 L 605.35714285714286 528.72142857142857 L 608.03571428571429 531.6 L 610.71428571428571 534.47857142857143 L 613.39285714285714 537.35714285714286 L 616.07142857142857 540.23571428571429 L 618.75 543.11428571428571 L 621.42857142857143 546.0 L 624.10714285714286 548.87857142857143 L 626.78571428571429 551.75714285714286 L 629.46428571428571 554.63571428571429 L 632.14285714285714 557.51428571428571 L 634.82142857142857 560.39285714285714 L 637.5 563.27142857142857 L 640.17857142857143 566.15 L 642.85714285714286 569.02857142857143 L 645.53571428571429 571.90714285714286 L 648.21428571428571 574.78571428571429 L 650.89285714285714 577.66428571428571 L 653.57142857142857 580.54285714285714 L 656.25 583.42142857142857 L 658.92857142857143 586.3 L 661.60714285714286 589.17857142857143 L 664.28571428571429 592.05714285714286 L 666.96428571428571 594.93571428571429 L 669.64285714285714 597.81428571428571 L 672.32142857142857 600.69285714285714 L 675 603.57142857142857 L 677.67857142857143 606.45 L 680.35714285714286 609.32857142857143 L 683.03571428571429 612.20714285714286 L 685.71428571428571 615.08571428571429 L 688.39285714285714 617.96428571428571 L 691.07142857142857 620.84285714285714 L 693.75 623.72142857142857 L 696.42857142857143 626.6 L 699.10714285714286 629.47857142857143 L 701.78571428571429 632.35714285714286 L 704.46428571428571 635.23571428571429 L 707.14285714285714 638.11428571428571 L 709.82142857142857 641.0 L 712.5 643.87857142857143 L 715.17857142857143 646.75714285714286 L 717.85714285714286 650.63571428571429 L 720.53571428571429 654.51428571428571 L 723.21428571428571 658.39285714285714 L 725.89285714285714 662.27142857142857 L 728.57142857142857 666.15 L 731.25 670.02857142857143 L 733.92857142857143 673.90714285714286 L 736.60714285714286 677.78571428571429 L 739.28571428571429 681.66428571428571 L 741.96428571428571 685.54285714285714 L 744.64285714285714 689.42142857142857 L 747.32142857142857 693.3 L 750 697.17857142857143 L 752.67857142857143 701.05714285714286 L 755.35714285714286 708.73571428571429 L 758.03571428571429 716.41428571428571 L 760.71428571428571 724.09285714285714 L 763.39285714285714 731.77142857142857 L 766.07142857142857 739.45 L 768.75 747.12857142857143 L 771.42857142857143 754.78571428571429 L 774.10714285714286 762.44285714285714 L 776.78571428571429 770.1 L 779.46428571428571 777.75714285714286 L 782.14285714285714 785.41428571428571 L 784.82142857142857 793.07142857142857 L 787.5 800.72857142857143 L 790.17857142857143 808.38571428571429 L 792.85714285714286 816.04285714285714 L 795.53571428571429 823.7 L 798.21428571428571 831.35714285714286 L 800.89285714285714 839.01428571428571 L 803.57142857142857 846.67142857142857 L 806.25 854.32857142857143 L 808.92857142857143 861.98571428571429 L 811.60714285714286 869.64285714285714 L 814.28571428571429 877.3 L 816.96428571428571 884.95714285714286 L 819.64285714285714 892.61428571428571 L 822.32142857142857 900.27142857142857 L 825 907.92857142857143 L 827.67857142857143 915.58571428571429 L 830.35714285714286 923.24285714285714 L 833.03571428571429 930.9 L 835.71428571428571 938.55714285714286 L 838.39285714285714 946.21428571428571 L 841.07142857142857 953.87142857142857 L 843.75 961.52857142857143 L 846.42857142857143 969.18571428571429 L 849.10714285714286 976.84285714285714 L 851.78571428571429 984.5 L 854.46428571428571 992.15714285714286 L 857.14285714285714 1000.0 L 859.82142857142857 1007.8571428571428 L 862.5 1015.7142857142857 L 865.17857142857143 1023.5714285714286 L 867.85714285714286 1031.4285714285714 L 870.53571428571429 1039.2857142857142 L 873.21428571428571 1047.1428571428571 L 875.89285714285714 1055.0 L 878.57142857142857 1062.8571428571428 L 881.25 1070.7142857142857 L 883.92857142857143 1078.5714285714286 L 886.60714285714286 1086.4285714285714 L 889.28571428571429 1094.2857142857142 L 891.96428571428571 1102.1428571428571 L 894.64285714285714 1110.0 L 897.32142857142857 1117.8571428571428 L 900 1125.7142857142857 L 902.67857142857143 1133.5714285714286 L 905.35714285714286 1141.4285714285714 L 908.03571428571429 1149.2857142857142 L 910.71428571428571 1157.1428571428571 L 913.39285714285714 1165.0 L 916.07142857142857 1172.8571428571428 L 918.75 1180.7142857142857 L 921.42857142857143 1188.5714285714286 L 924.10714285714286 1196.4285714285714 L 926.78571428571429 1204.2857142857142 L 929.46428571428571 1212.1428571428571 L 932.14285714285714 1220.0 L 934.82142857142857 1227.8571428571428 L 937.5 1235.7142857142857 L 940.17857142857143 1243.57
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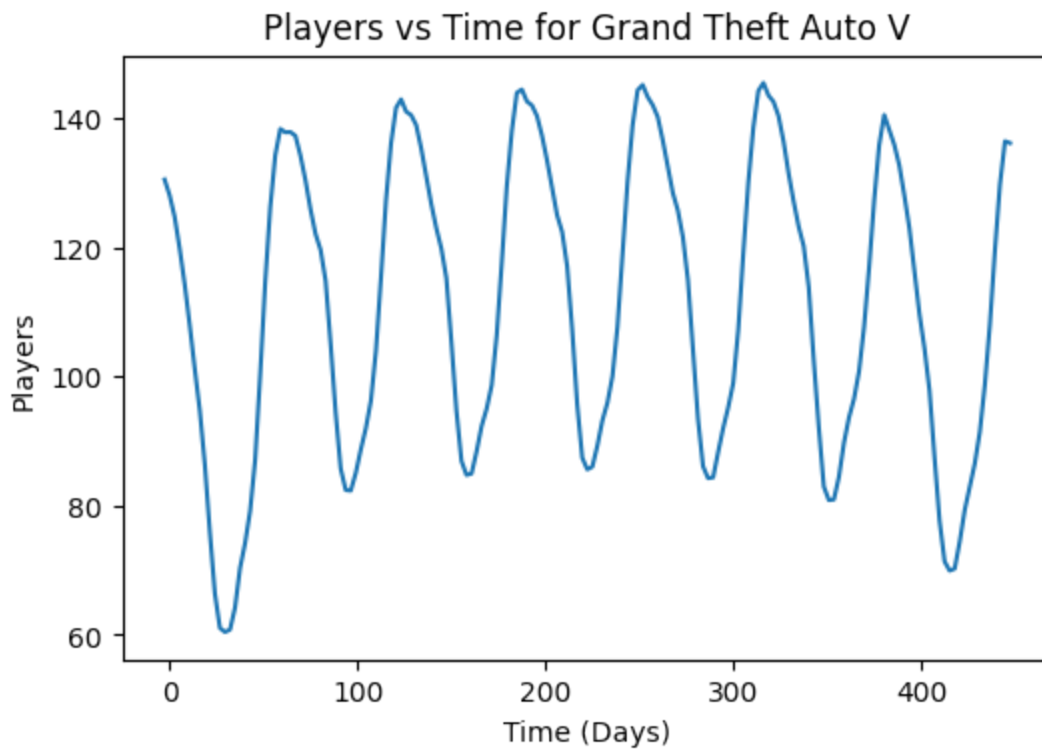


In [145...

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name = "Rust"
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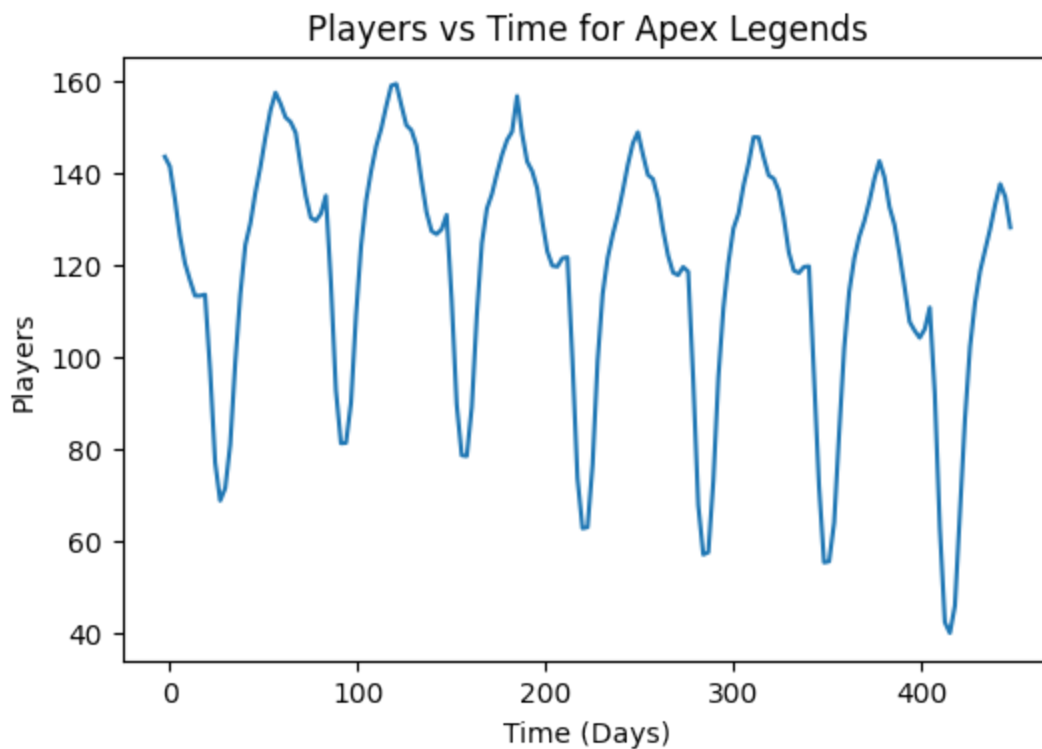
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2465.33828571428571 L 416.33714285714285 2482.46428571428571 L 419.00571428571429 2500.59090909090909 L 421.67428571428571 2517.71428571428571 L 424.34285714285714 2534.83828571428571 L 427.01142857142858 2551.96428571428571 L 429.68 2569.09090909090909 L 432.34857142857143 2586.21428571428571 L 435.01714285714285 2603.33828571428571 L 437.68571428571429 2620.46428571428571 L 440.35428571428571 2637.59090909090909 L 443.02285714285714 2654.71428571428571 L 445.69142857142858 2671.83828571428571 L 448.36 2688.96428571428571 L 451.02857142857143 2706.09090909090909 L 453.69714285714285 2723.21428571428571 L 456.36571428571429 2740.33828571428571 L 459.03428571428571 2757.46428571428571 L 461.70285714285714 2774.59090909090909 L 464.37142857142858 2791.71428571428571 L 467.04 2808.83828571428571 L 469.70857142857143 2825.96428571428571 L 472.37714285714285 2843.09090909090909 L 475.04571428571429 2860.21428571428571 L 477.71428571428571 2877.33828571428571 L 480.38285714285714 2894.46428571428571 L 483.05142857142858 2911.59090909090909 L 485.72 2928.71428571428571 L 488.38857142857143 2945.83828571428571 L 491.05714285714285 2962.96428571428571 L 493.72571428571429 2980.09090909090909 L 496.39428571428571 2997.21428571428571 L 499.06285714285714 3014.33828571428571 L 501.73142857142858 3031.46428571428571 L 504.4 3048.59090909090909 L 507.06857142857143 3065.71428571428571 L 509.73714285714285 3082.83828571428571 L 512.40571428571429 3100.96428571428571 L 515.07428571428571 3118.09090909090909 L 517.74285714285714 3135.21428571428571 L 520.41142857142858 3152.33828571428571 L 523.08 3169.46428571428571 L 525.74857142857143 3186.59090909090909 L 528.41714285714285 3203.71428571428571 L 531.08571428571429 3220.83828571428571 L 533.75428571428571 3237.96428571428571 L 536.42285714285714 3255.09090909090909 L 539.09142857142858 3272.21428571428571 L 541.76 3289.33828571428571 L 544.42857142857143 3306.46428571428571 L 547.09714285714285 3323.59090909090909 L 549.76571428571429 3340.71428571428571 L 552.43428571428571 3357.83828571428571 L 555.10285714285714 3374.96428571428571 L 557.77142857142858 3392.09090909090909 L 560.44 3409.21428571428571 L 563.10857142857143 3426.33828571428571 L 565.77714285714285 3443.46428571428571 L 568.44571428571429 3460.59090909090909 L 571.11428571428571 3477.71428571428571 L 573.78285714285714 3494.83828571428571 L 576.45142857142858 3511.96428571428571 L 579.12 3529.09090909090909 L 581.78857142857143 3546.21428571428571 L 584.45714285714285 3563.33828571428571 L 587.12571428571429 3580.46428571428571 L 589.79428571428571 3597.59090909090909 L 592.46285714285714 3614.71428571428571 L 595.13142857142858 3631.83828571428571 L 597.8 3648.96428571428571 L 600.46857142857143 3666.09090909090909 L 603.13714285714285 3683.21428571428571 L 605.80571428571429 3700.33828571428571 L 608.47428571428571 3717.46428571428571 L 611.14285714285714 3734.59090909090909 L 613.81142857142858 3751.71428571428571 L 616.48 3768.83828571428571 L 619.14857142857143 3785.96428571428571 L 621.81714285714285 3803.09090909090909 L 624.48571428571429 3820.21428571428571 L 627.15428571428571 3837.33828571428571 L 629.82285714285714 3854.46428571428571 L 632.49142857142858 3871.59090909090909 L 635.16 3888.71428571428571 L 637.82857142857143 3905.83828571428571 L 640.49714285714285 3922.96428571428571 L 643.16571428571429 3940.09090909090909 L 645.83428571428571 3957.21428571428571 L 648.50285714285714 3974.33828571428571 L 651.17142857142858 3991.46428571428571 L 653.84 4008.59090909090909 L 656.50857142857143 4025.71428571428571 L 659.17714285714285 4042.83828571428571 L 661.84571428571429 4059.96428571428571 L 664.51428571428571 4077.09090909090909 L 667.18285714285714 4094.21428571428571 L 669.85142857142858 4111.33828571428571 L 672.52 4128.46428571428571 L 675.18857142857143 4145.59090909090909 L 677.85714285714285 4162.71428571428571 L 680.52571428571429 4179.83828571428571 L 683.19428571428571 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819.29142857142858 5070.33828571428571 L 821.96 5087.46428571428571 L 824.62857142857143 5104.59090909090909 L 827.29714285714285 5121.71428571428571 L 830.0 5138.83828571428571 L 832.66857142857143 5155.96428571428571 L 835.33714285714285 5173.09090909090909 L 838.00571428571429 5190.21428571428571 L 840.67428571428571 5207.33828571428571 L 843.34285714285714 5224.46428571428571 L 846.01142857142858 5241.59090909090909 L 848.68 5258.71428571428571 L 851.34857142857143 5275.83828571428571 L 854.01714285714285 5292.96428571428571 L 856.68571428571429 5310.09090909090909 L 859.35428571428571 5327.21428571428571 L 862.02285714285714 5344.33828571428571 L 864.69142857142858 5361.46428571428571 L 867.36 5378.59090
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In [148...

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data = "M -2.6785714285714 143.43813 L 0 141.32052 L 2.6785714285714 134.29944 L 5.357142857142857 127.14285714285714 L 8.035714285714286 119.99999999999999 L 10.714285714285714 112.85714285714286 L 13.392857142857143 105.71428571428571 L 16.07142857142857 98.57142857142857 L 18.75 91.42857142857143 L 21.42857142857143 84.28571428571429 L 24.10714285714286 77.14285714285714 L 26.785714285714285 70 L 29.464285714285715 62.857142857142854 L 32.142857142857146 55.714285714285715 L 34.82142857142857 48.57142857142857 L 37.5 41.42857142857143 L 40.17857142857143 34.285714285714285 L 42.857142857142854 27.142857142857143 L 45.535714285714286 20 L 48.214285714285715 12.857142857142857 L 50.892857142857146 5.714285714285714 L 53.57142857142857 0 L 56.25 0 L 58.92857142857143 0 L 61.607142857142854 0 L 64.28571428571429 0 L 66.964285714285715 0 L 69.642857142857146 0 L 72.32142857142857 0 L 75 0 L 77.67857142857143 0 L 80.35714285714286 0 L 83.03571428571429 0 L 85.71428571428571 0 L 88.39285714285714 0 L 91.07142857142857 0 L 93.75 0 L 96.42857142857143 0 L 99.10714285714286 0 L 101.78571428571429 0 L 104.46428571428571 0 L 107.14285714285714 0 L 109.82142857142857 0 L 112.5 0 L 115.17857142857143 0 L 117.85714285714286 0 L 120.53571428571429 0 L 123.21428571428571 0 L 125.89285714285714 0 L 128.57142857142857 0 L 131.25 0 L 133.92857142857143 0 L 136.60714285714286 0 L 139.28571428571429 0 L 141.96428571428571 0 L 144.64285714285714 0 L 147.32142857142857 0 L 150 0 L 152.67857142857143 0 L 155.35714285714286 0 L 158.03571428571429 0 L 160.71428571428571 0 L 163.39285714285714 0 L 166.07142857142857 0 L 168.75 0 L 171.42857142857143 0 L 174.10714285714286 0 L 176.78571428571429 0 L 179.46428571428571 0 L 182.14285714285714 0 L 184.82142857142857 0 L 187.5 0 L 190.17857142857143 0 L 192.85714285714286 0 L 195.53571428571429 0 L 198.21428571428571 0 L 200.89285714285714 0 L 203.57142857142857 0 L 206.25 0 L 208.92857142857143 0 L 211.60714285714286 0 L 214.28571428571429 0 L 216.96428571428571 0 L 219.64285714285714 0 L 222.32142857142857 0 L 225 0 L 227.67857142857143 0 L 230.35714285714286 0 L 233.03571428571429 0 L 235.71428571428571 0 L 238.39285714285714 0 L 241.07142857142857 0 L 243.75 0 L 246.42857142857143 0 L 249.10714285714286 0 L 251.78571428571429 0 L 254.46428571428571 0 L 257.14285714285714 0 L 259.82142857142857 0 L 262.5 0 L 265.17857142857143 0 L 267.85714285714286 0 L 270.53571428571429 0 L 273.21428571428571 0 L 275.89285714285714 0 L 278.57142857142857 0 L 281.25 0 L 283.92857142857143 0 L 286.60714285714286 0 L 289.28571428571429 0 L 291.96428571428571 0 L 294.64285714285714 0 L 297.32142857142857 0 L 300 0 L 302.67857142857143 0 L 305.35714285714286 0 L 308.03571428571429 0 L 310.71428571428571 0 L 313.39285714285714 0 L 316.07142857142857 0 L 318.75 0 L 321.42857142857143 0 L 324.10714285714286 0 L 326.78571428571429 0 L 329.46428571428571 0 L 332.14285714285714 0 L 334.82142857142857 0 L 337.5 0 L 340.17857142857143 0 L 342.85714285714286 0 L 345.53571428571429 0 L 348.21428571428571 0 L 350.89285714285714 0 L 353.57142857142857 0 L 356.25 0 L 358.92857142857143 0 L 361.60714285714286 0 L 364.28571428571429 0 L 366.96428571428571 0 L 369.64285714285714 0 L 372.32142857142857 0 L 375 0 L 377.67857142857143 0 L 380.35714285714286 0 L 383.03571428571429 0 L 385.71428571428571 0 L 388.39285714285714 0 L 391.07142857142857 0 L 393.75 0 L 396.42857142857143 0 L 399.10714285714286 0 L 401.78571428571429 0 L 404.46428571428571 0 L 407.14285714285714 0 L 409.82142857142857 0 L 412.5 0 L 415.17857142857143 0 L 417.85714285714286 0 L 420.53571428571429 0 L 423.21428571428571 0 L 425.89285714285714 0 L 428.57142857142857 0 L 431.25 0 L 433.92857142857143 0 L 436.60714285714286 0 L 439.28571428571429 0 L 441.96428571428571 0 L 444.64285714285714 0 L 447.32142857142857 0 L 450 0"
x, y = extract_players_vs_time(data)
plt.show()
games = games.append({"Name": name, "Players Vs Time": y}, ignore_index=True)
plot_players_vs_time(x, y, name)
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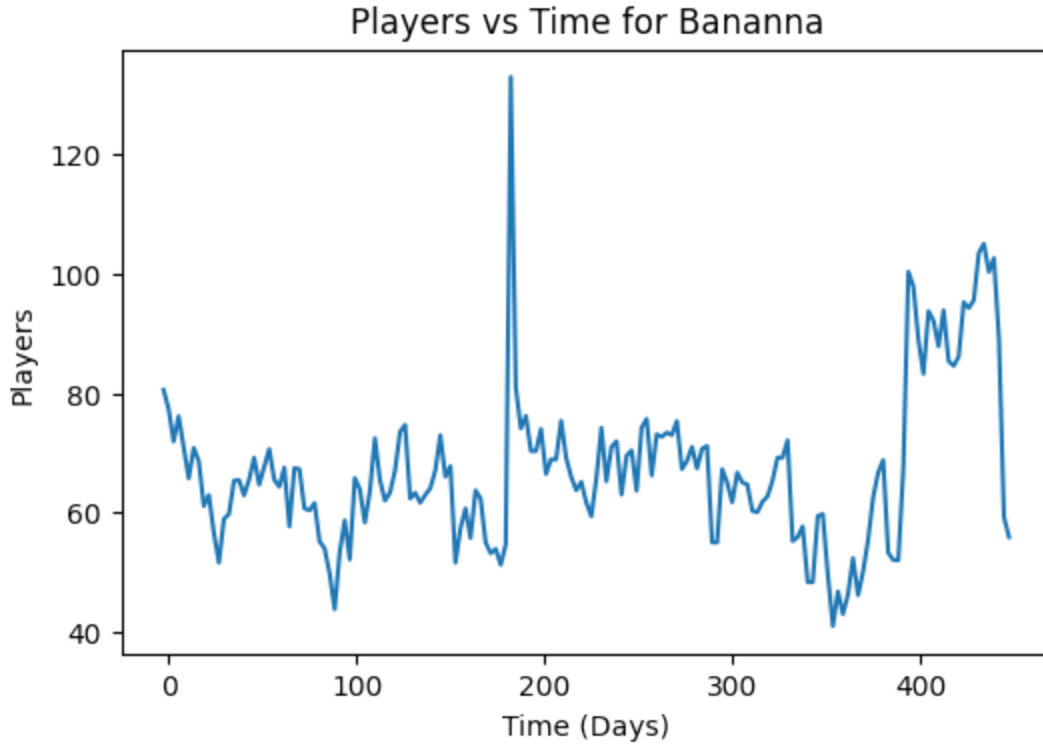
len(x) = 169



In [149...

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data = "M -2.6785714285714 80.57891 L 0 77.36387 L 2.6785714285714 71.913865 L 5.357142857142857"
x, y = extract_players_vs_time(data)
plt.show()
games = games.append({"Name": name, "Players Vs Time": y}, ignore_index=True)
plot_players_vs_time(x, y, name)
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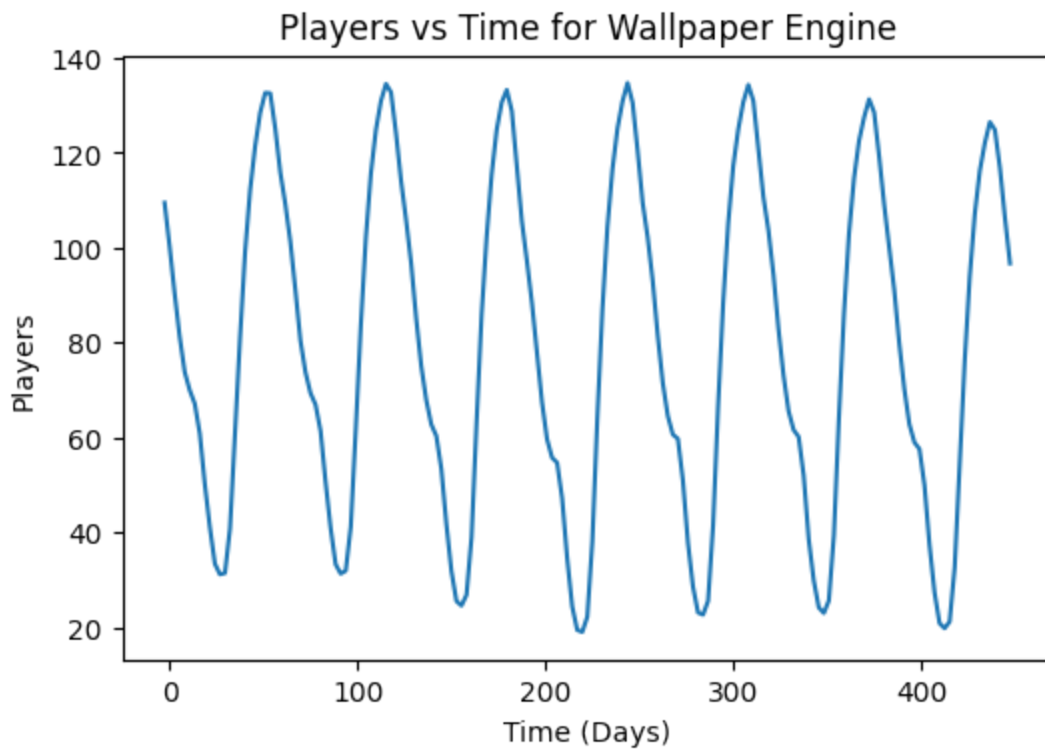
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len(x) = 169
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In [150...

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name = "Wallpaper Engine"  
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x, y = extract_players_vs_time(data)  
plt.show()  
games = games.append({"Name": name, "Players Vs Time": y}, ignore_index=True)  
plot_players_vs_time(x, y, name)
```

```
len(x) = 169
```



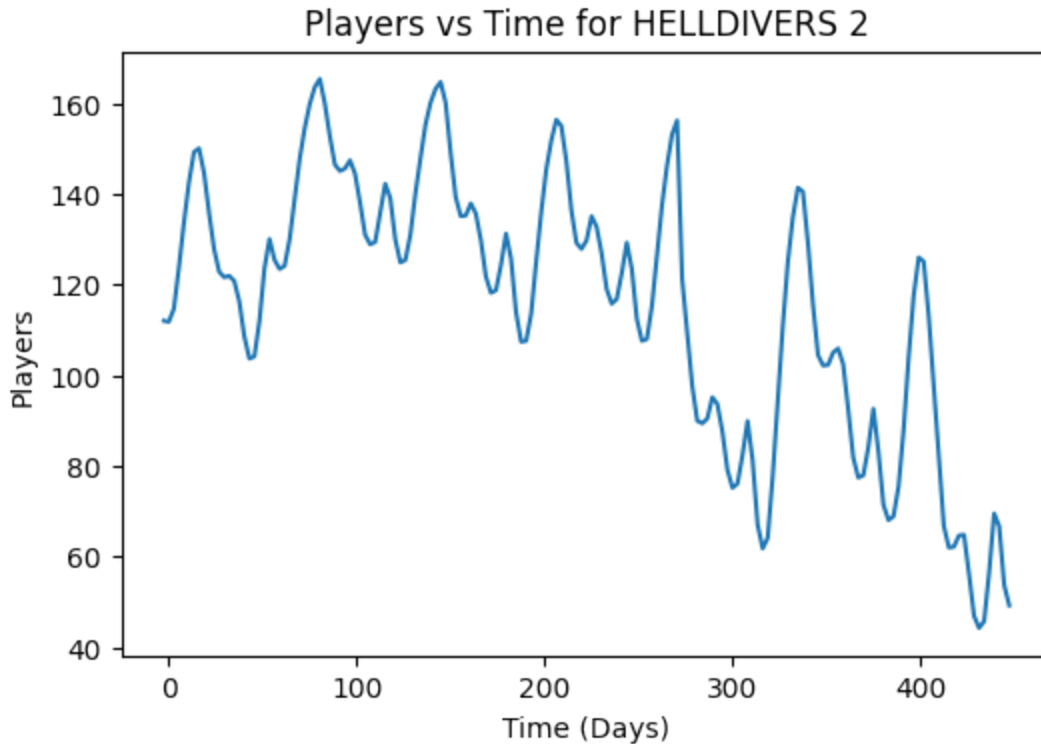
In [151...

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name = "Stardew Valley"
data = "M -2.6785714285714 107.92848 L 0 106.3722 L 2.6785714285714 104.06352 L 5.357142857142857 102.14285714285714 L 8.042857142857143 100.21428571428571 L 10.728571428571429 98.28571428571429 L 13.414285714285714 96.35714285714286 L 16.100000000000001 94.42857142857143 L 18.785714285714285 92.50000000000001 L 21.47142857142857 90.57142857142857 L 24.157142857142858 88.64285714285714 L 26.842857142857142 86.71428571428571 L 29.528571428571427 84.78571428571428 L 32.214285714285715 82.85714285714285 L 34.900000000000004 80.92857142857143 L 37.585714285714286 79.00000000000001 L 40.27142857142857 77.07142857142857 L 42.957142857142854 75.14285714285714 L 45.642857142857146 73.21428571428571 L 48.32857142857143 71.28571428571428 L 51.014285714285715 69.35714285714285 L 53.700000000000004 67.42857142857143 L 56.385714285714286 65.50000000000001 L 59.07142857142857 63.57142857142857 L 61.757142857142854 61.64285714285714 L 64.442857142857146 59.71428571428571 L 67.12857142857143 57.78571428571428 L 69.814285714285715 55.85714285714285 L 72.50000000000001 53.92857142857143 L 75.185714285714286 52.00000000000001 L 77.87142857142857 50.07142857142857 L 80.557142857142854 48.14285714285714 L 83.242857142857146 46.21428571428571 L 85.92857142857143 44.28571428571428 L 88.614285714285715 42.35714285714285 L 91.30000000000001 40.42857142857143 L 93.985714285714286 38.50000000000001 L 96.67142857142857 36.57142857142857 L 99.357142857142854 34.64285714285714 L 102.042857142857146 32.71428571428571 L 104.72857142857143 30.78571428571428 L 107.414285714285715 28.85714285714285 L 110.10000000000001 26.92857142857143 L 112.785714285714286 25.00000000000001 L 115.47142857142857 23.07142857142857 L 118.157142857142854 21.14285714285714 L 120.842857142857146 19.21428571428571 L 123.52857142857143 17.28571428571428 L 126.214285714285715 15.35714285714285 L 128.90000000000001 13.42857142857143 L 131.585714285714286 11.50000000000001 L 134.27142857142857 9.57142857142857 L 136.957142857142854 7.64285714285714 L 139.642857142857146 5.71428571428571 L 142.32857142857143 3.78571428571428 L 145.014285714285715 1.85714285714285 L 147.70000000000001 0.00000000000000 L 150.385714285714286 -1.85714285714285 L 153.07142857142857 -3.78571428571428 L 155.757142857142854 -5.71428571428571 L 158.442857142857146 -7.64285714285714 L 161.12857142857143 -9.57142857142857 L 163.814285714285715 -11.50000000000001 L 166.50000000000001 -13.42857142857143 L 169.185714285714286 -15.35714285714285 L 171.87142857142857 -17.28571428571428 L 174.557142857142854 -19.21428571428571 L 177.242857142857146 -21.14285714285714 L 179.92857142857143 -23.07142857142857 L 182.614285714285715 -25.00000000000001 L 185.30000000000001 -26.92857142857143 L 187.985714285714286 -28.85714285714285 L 190.67142857142857 -30.78571428571428 L 193.357142857142854 -32.71428571428571 L 196.042857142857146 -34.64285714285714 L 198.72857142857143 -36.57142857142857 L 201.414285714285715 -38.50000000000001 L 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-85.35714285714285 L 271.242857142857146 -87.21428571428571 L 273.92857142857143 -89.07142857142857 L 276.614285714285715 -90.92857142857143 L 279.30000000000001 -92.78571428571428 L 281.985714285714286 -94.64285714285714 L 284.67142857142857 -96.50000000000001 L 287.357142857142854 -98.35714285714285 L 290.042857142857146 -100.21428571428571 L 292.72857142857143 -102.07142857142857 L 295.414285714285715 -103.92857142857143 L 298.10000000000001 -105.78571428571428 L 300.785714285714286 -107.64285714285714 L 303.47142857142857 -109.50000000000001 L 306.157142857142854 -111.35714285714285 L 308.842857142857146 -113.21428571428571 L 311.52857142857143 -115.07142857142857 L 314.214285714285715 -116.92857142857143 L 316.90000000000001 -118.78571428571428 L 319.585714285714286 -120.64285714285714 L 322.27142857142857 -122.50000000000001 L 324.957142857142854 -124.35714285714285 L 327.642857142857146 -126.21428571428571 L 330.32857142857143 -128.07142857142857 L 333.014285714285715 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```


In [152...

```
name = "HELLDIVERS 2"  
data = "M -2.6785714285714 111.984048 L 0 111.762288 L 2.6785714285714 114.640416 L 5.3  
x, y = extract_players_vs_time(data)  
plt.show()  
games = games.append({"Name": name, "Players Vs Time": y}, ignore_index=True)  
plot_players_vs_time(x, y, name)
```

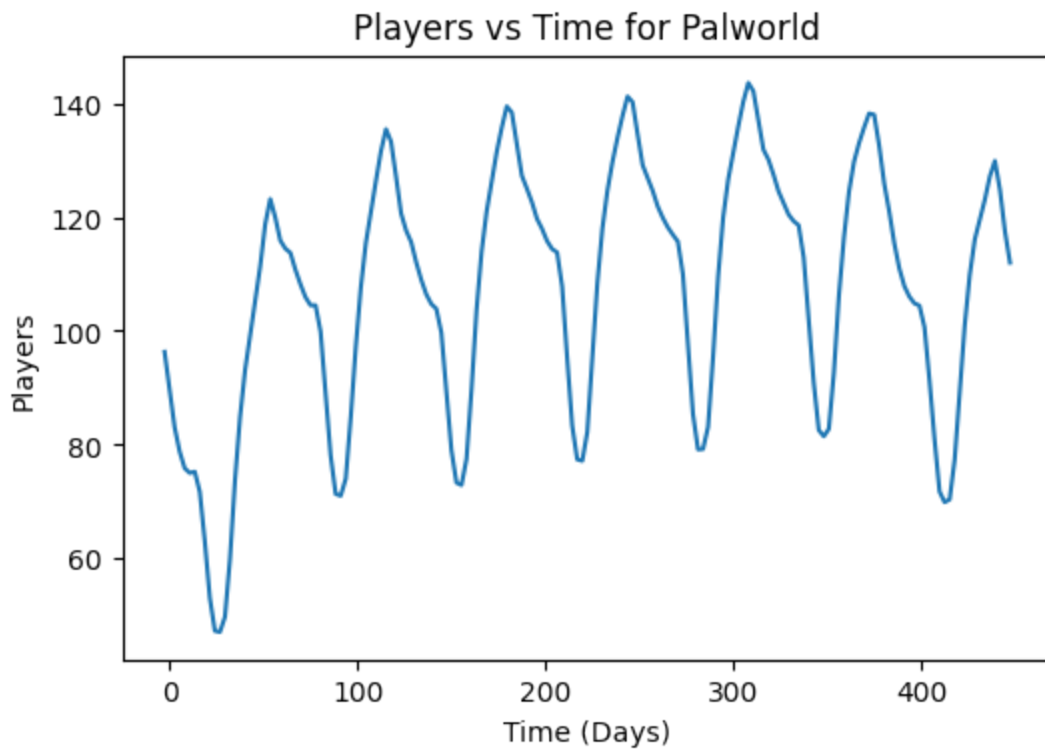
len(x) = 169



In [153...

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name = "Palworld"  
data = "M -2.6785714285714 96.33228 L 0 89.84184 L 2.6785714285714 83.07156 L 5.3571428  
x, y = extract_players_vs_time(data)  
plt.show()  
games = games.append({"Name": name, "Players Vs Time": y}, ignore_index=True)  
plot_players_vs_time(x, y, name)
```

len(x) = 169



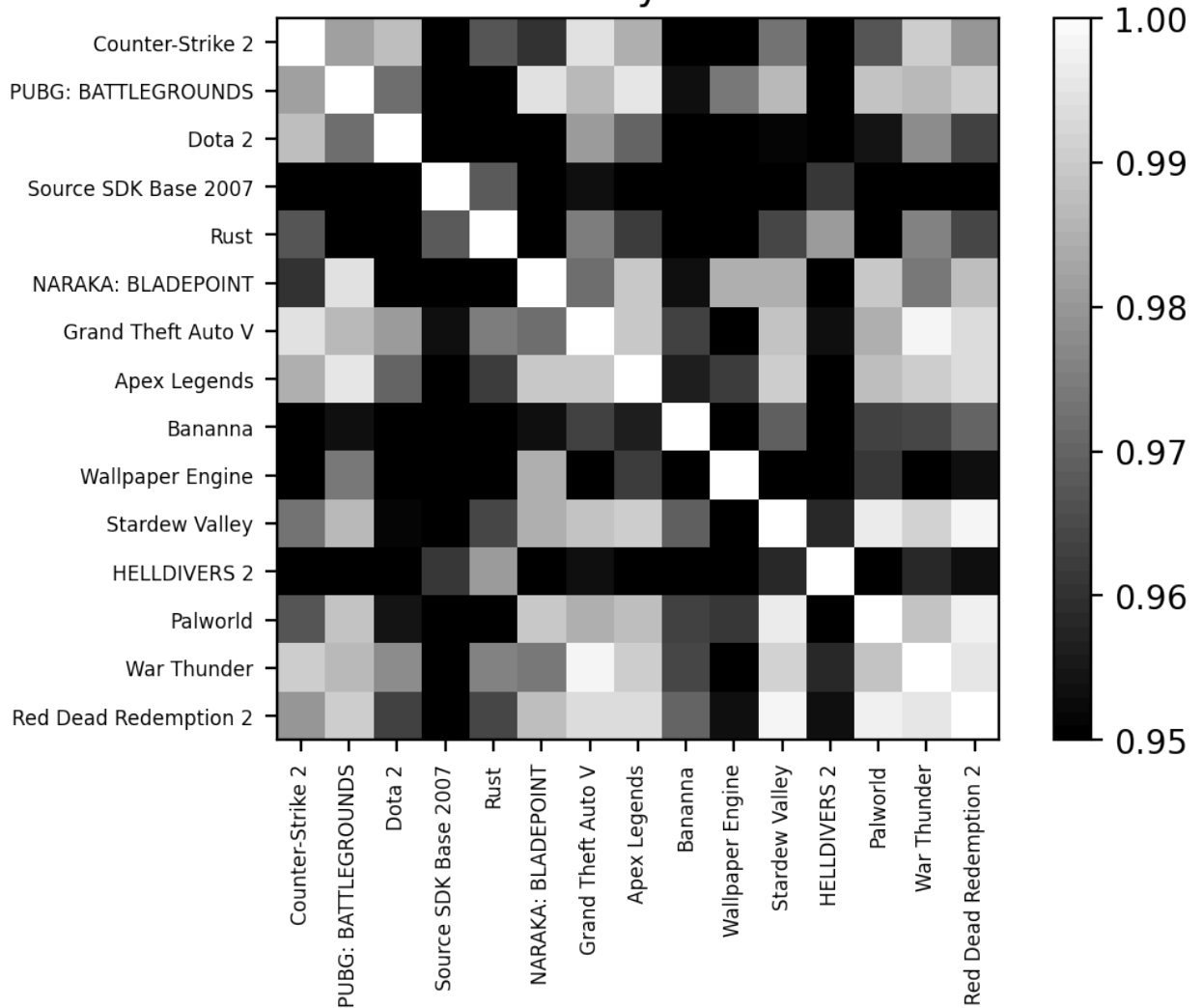
In [154...

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name = "War Thunder"
data = "M -2.6785714285714 129.7494 L 0 128.41224 L 2.6785714285714 126.92988 L 5.357142857142857 124.85714285714286 L 8.035714285714286 122.78571428571428 L 10.714285714285714 120.71428571428571 L 13.392857142857143 118.64285714285714 L 16.07142857142857 116.57142857142857 L 18.75 114.5 L 21.42857142857143 112.42857142857143 L 24.107142857142858 110.35714285714286 L 26.785714285714286 108.28571428571428 L 29.464285714285715 106.21428571428571 L 32.142857142857146 104.14285714285714 L 34.82142857142857 102.07142857142857 L 37.5 100 L 40.17857142857143 97.92857142857143 L 42.857142857142854 95.85714285714286 L 45.535714285714286 93.78571428571428 L 48.21428571428571 91.71428571428571 L 50.89285714285714 89.64285714285714 L 53.57142857142857 87.57142857142857 L 56.25 85.5 L 58.92857142857143 83.42857142857143 L 61.607142857142858 81.35714285714286 L 64.285714285714286 79.28571428571428 L 66.96428571428571 77.21428571428571 L 69.64285714285714 75.14285714285714 L 72.32142857142857 73.07142857142857 L 75 71 L 77.67857142857143 68.92857142857143 L 80.35714285714286 66.85714285714286 L 83.03571428571428 64.78571428571428 L 85.71428571428571 62.71428571428571 L 88.39285714285714 60.64285714285714 L 91.07142857142857 58.57142857142857 L 93.75 56.5 L 96.42857142857143 54.42857142857143 L 99.10714285714286 52.35714285714286 L 101.78571428571428 50.28571428571428 L 104.46428571428571 48.21428571428571 L 107.14285714285714 46.14285714285714 L 109.82142857142857 44.07142857142857 L 112.5 42 L 115.17857142857143 39.92857142857143 L 117.85714285714286 37.85714285714286 L 120.53571428571428 35.78571428571428 L 123.21428571428571 33.71428571428571 L 125.89285714285714 31.64285714285714 L 128.57142857142857 29.57142857142857 L 131.25 27.5 L 133.92857142857143 25.42857142857143 L 136.60714285714286 23.35714285714286 L 139.28571428571428 21.285714285714286 L 141.96428571428571 19.21428571428571 L 144.64285714285714 17.14285714285714 L 147.32142857142857 15.07142857142857 L 150 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590.84285714285714 -721.95836 L 593.52857142857143 -725.76247 L 596.21428571428571 -729.56658 L 598.90000000000001 -733.37069 L 601.58571428571429 -737.17480 L 604.27142857142857 -740.97891 L 606.95714285714286 -744.78302 L 609.64285714285714 -748.58713 L 612.32857142857143 -752.39124 L 615.01428571428571 -756.19535 L 617.70000000000001 -759.99946 L 620.38571428571429 -763.80357 L 623.07142857142857 -767.60768 L 625.75714285714286 -771.41179 L 628.44285714285714 -775.21590 L 631.12857142857143 -779.02001 L 633.81428571428571 -782.82412 L 636.50000000000001 -786.62823 L 639.18571428571429 -790.43234 L 641.87142857142857 -794.23645 L 644.55714285714286 -798.04056 L 647.24285714285714 -801.84467 L 649.92857142857143 -805.64878 L 652.61428571428571 -809.45289 L 655.30000000000001 -813.25700 L 657.98571428571429 -817.06111 L 660.67142857142857 -820.86522 L 663.35714285714286 -824.66933 L 666.04285714285714 -828.47344 L 668.72857142857143 -832.27755 L 671.41428571428571 -836.08166 L 674.10000000000001 -839.88577 L 676.78571428571429 -843.68988 L 679.47142857142857 -847.49399 L 682.15714285714286 -851.29810 L 684.84285714285714 -855.10221 L 687.52857142857143 -858.90632 L 690.21428571428571 -862.71043 L 692.90000000000001 -866.51454 L 695.58571428571429 -870.31865 L 698.27142857142857 -874.12276 L 700.95714285714286 -877.92687 L 703.64285714285714 -881.73098 L 706.32857142857143 -885.53509 L 709.01428571428571 -889.33920 L 711.70000000000001 -893.14331 L 714.38571428571429 -896.94742 L 717.07142857142857 -900.75153 L 719.75714285714286 -904.55564 L 722.44285714285714 -908.35975 L 725.12857142857143 -912.16386 L 727.81428571428571 -915.96797 L 730.50000000000001 -919.77208 L 733.18571428571429 -923.57619 L 735.87142857142857 -927.38030 L 738.55714285714286 -931.18441 L 741.24285714285714 -934.98852 L 743.92857142857143 -938.79263 L 746.61428571428571 -942.59674 L 749.30000000000001 -946.40085 L 751.98571428571429 -950.20496 L 754.67142857142857 -954.00907 L 757.35714285714286 -957.81318 L 760.04285714285714 -961.61729 L 762.72857142857143 -965.42140 L 765.41428571428571 -969.22551 L 768.10000000000001 -973.02962 L 770.78571428571429 -976.83373 L 773.47142857142857 -980.63784 L 776.15714285714286 -984.44195 L 778.84285714285714 -988.24606 L 781.52857142857143 -992.05017 L 784.21428571428571 -995.85428 L 786.90000000000001 -999.65839 L 789.58571428571429 -1003.46250 L 792.27142857142857 -1007.26661 L 794.95714285714286 -1011.07072 L 797.64285714285714 -1014.87483 L 800.32857142857143 -1018.67894 L 803.01428571428571 -1022.48305 L 805.70000000000001 -1026.28716 L 808.38571428571429 -1030.09127 L 811.07142857142857 -1033.89538 L 813.75714285714286 -1037.69949 L 816.44285714285714 -1041.50360 L 819.12857142857143 -1045.30771 L 821.81428571428571 -1049.11182 L 824.50000000000001 -1052.91593 L 827.18571428571429 -1056.72004 L 829.87142857142857 -1060.52415 L 832.55714285714286 -1064.32826 L 835.24285714285714 -1068.13237 L 837.92857142857143 -1071.93648 L 840.61428571428571 -1075.74059 L 843.30000000000001 -1079.54470 L 845.98571428571429 -1083.34881 L 848.67142857142857 -1087.15292 L 851.35714285714286 -1090.95703 L 854.04285714285714 -1094.76114 L 856.72857142857143 -1098.56525 L 859.41428571428571 -1102.36936 L 862.10000000000001 -1106.17347 L 864.78571428571429 -1109.97758 L 867.47142857142857 -1113.78169 L 870.15714285714286 -1117.58580 L 872.84285714285714 -1121.38991 L 875.52857142857143 -1125.19402 L 878.21428571428571 -1128.99813 L 880.90000000000001 -1132.80224 L 883.58571428571429 -1136.60635 L 886.27142857142857 -1140.41046 L 888.95714285714286 -1144.21457 L 891.64285714285714 -1148.01868 L 894.32857142857143 -1151.82279 L 897.01428571428571 -1155.62690 L 899.70000000000001 -1159.43101 L 902.38571428571429 -1163.23512 L 905.07142857142857 -1167.03923 L 907.75714285714286 -1170.84334 L 910.44285714285714 -1174.64745 L 913.12857142857143 -1178.45156 L 915.81428571428571 -1182.25567 L 918.50000000000001 -1186.05978 L 921.18571428571429 -1189.86389 L 923.87142857142857 -1193.66800 L 926.55714285714286 -1197.47211 L 929.24285714285714 -1201.27622 L 931.92857142857143 -1205.08033 L 934.61428571428571 -1208.88444 L 937.30000000000001 -1212.68855 L 939.98571428571429 -1216.49266 L 942.67142857142857 -1220.29677 L 945.35714285714286 -1224.10088 L 948.04285714285714 -1227.90499 L 950.72857142857143 -1231.70910 L 953.41428571428571 -1235.51321 L 956.10000000000001 -1239.31732 L 958.78571428571429 -1243.12143 L 961.47142857142857 -1246.92554 L 964.15714285714286 -1250.72965 L 966.84285714285714 -1254.53376 L 969.52857142857143 -1258.33787 L 972.21428571428571 -1262.14198 L 974.90000000000001 -1265.94609 L 977.58571428571429 -1269.75020 L 980.27142857142857 -1273.55431 L 982.95714285714286 -1277.35842 L 985.64285714285714 -1281.16253 L 988.32857142857143 -1284.96664 L 991.01428571428571 -1288.77075 L 993.70000000000001 -1292.57486 L 996.38571428571429 -1296.37897 L 999.07142857142857 -1300.18308 L 1001.75714285714286 -1303.98719 L 1004.44285714285714 -1307.79130 L 1007.12857142857143 -1311.59541 L 1009.81428571428571 -1315.39952 L 1012.50000000000001 -1319.20363 L 1015.18571428571429 -1323.00774 L 1017.87142857142857 -1326.81185 L 1020.55714285714286 -1330.61596 L 1023.24285714285714 -1334.42007 L 1025.92857142857143 -1338.22418 L 1028.61428571428571 -1342.02829 L 1031.30000000000001 -1345.83240 L 1033.98571428571429 -1349.63651 L 1036.67142857142857 -1353.44062 L 1039.35714285714286 -1357.24473 L 1042.04285714285714 -1361.04884 L 1044.72857142857143 -1364.85295 L 1047.41428571428571 -1368.65706 L 1050.10000000000001 -1372.46117 L 1052.78571428571429 -1376.26528 L 1055.47142857142857 -1380.06939 L 1058.15714285714286 -1383.87350 L 1060.84285714285714 -1387.67761 L 1063.52857142857143 -1391.48172 L 1066.21428571428571 -1395.28583 L 1068.90000000000001 -1399.08994 L 1071.58571428571429 -1402.89405 L 1074.27142857142857 -1406.69816 L 1076.95714285714286 -1410.50227 L 1079.64285714285714 -1414.30638 L 1082.32857142857143 -1418.11049 L 1085.01428571428571 -1421.91460 L 1087.70000000000001 -1425.71871 L 1090.38571428571429 -1429.52282 L 1093.07142857142857 -14
```

Correlations Between Player Vs Time of 15 Games



Strongest Correlations

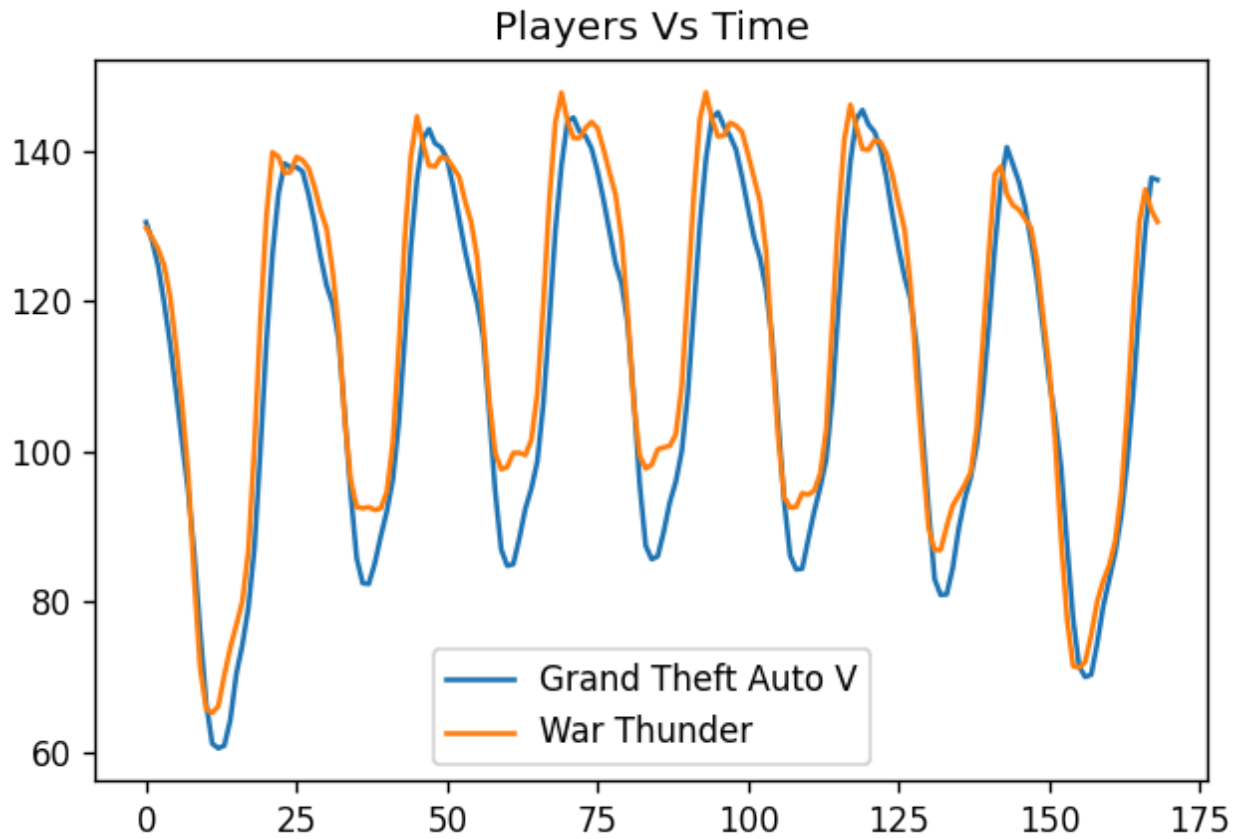
In [169...

```
def get_correlation(y1, y2):
    r = np.dot(y1,y2)/(np.sqrt(np.sum(y2*y2))*np.sqrt(np.sum(y1*y1)))
    return r

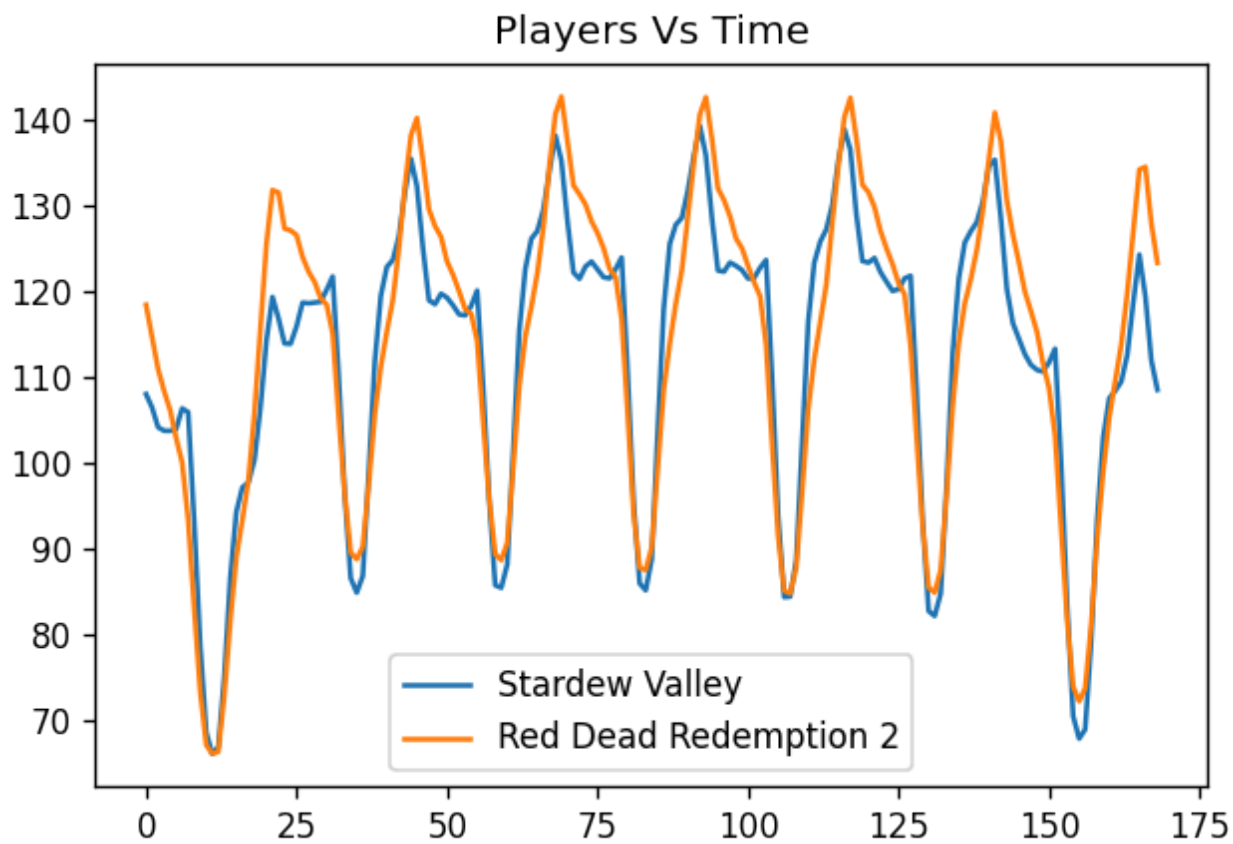
graphs = games[['Players Vs Time']][0].to_numpy()
names = games['Name'].values.tolist()
N = len(graphs[0])
correlations = np.zeros((len(graphs), len(graphs)))
for i in range(len(graphs)):
    for n in range(len(graphs)):
        correlations[i, n] = get_correlation(graphs[i][:N], graphs[n][:N])
for i in range(len(graphs)):
    correlations[i,i] = 0
for _ in range(5):
    print(f'{np.argmax(correlations) = }')
    i, n = np.unravel_index(np.argmax(correlations), np.shape(correlations))
    # i, n = np.argmax(correlations)
    plt.figure(dpi=120)
    plt.title('Players Vs Time')
    plt.plot(graphs[i], label=names[i])
```

```
plt.plot(graphs[n], label=names[n])  
plt.legend()  
plt.show()  
correlations[i,n], correlations[n,i] = 0, 0
```

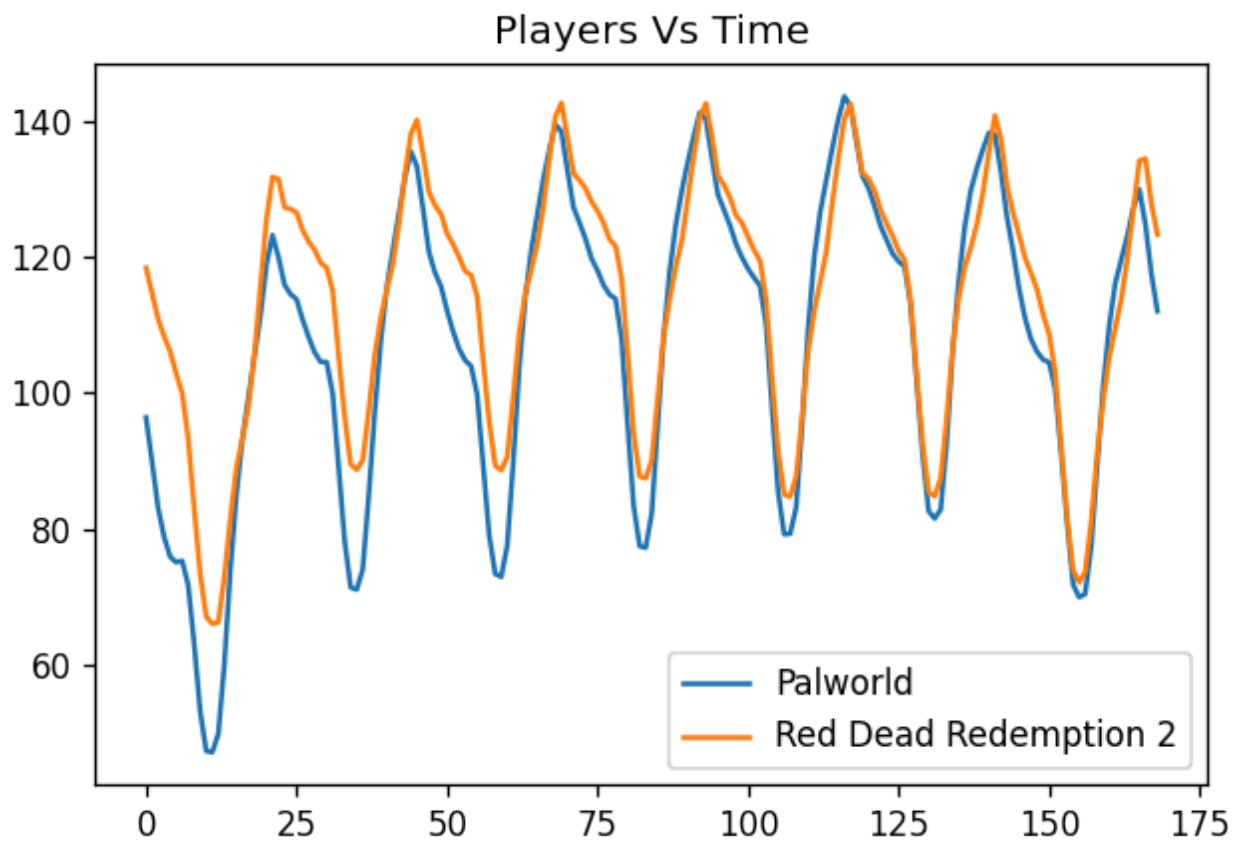
np.argmax(correlations) = 103



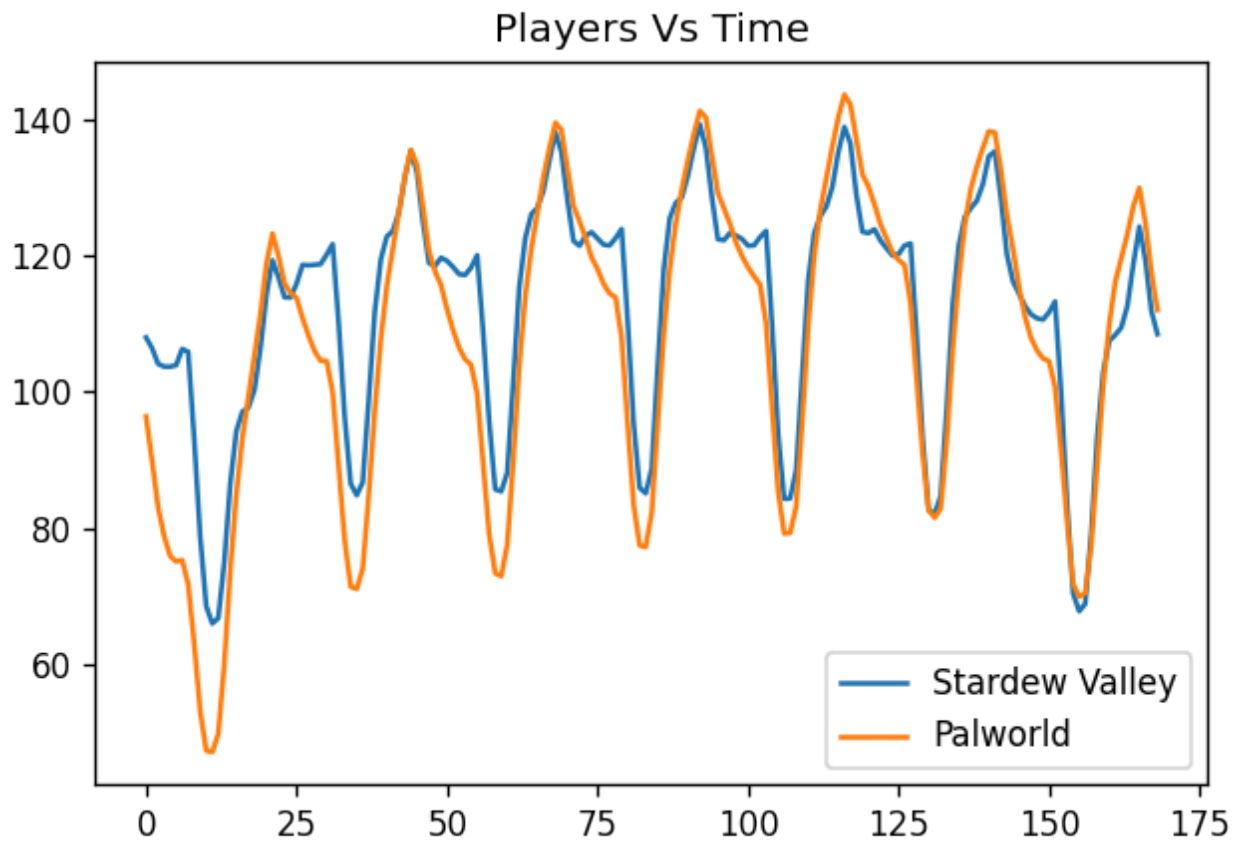
np.argmax(correlations) = 164



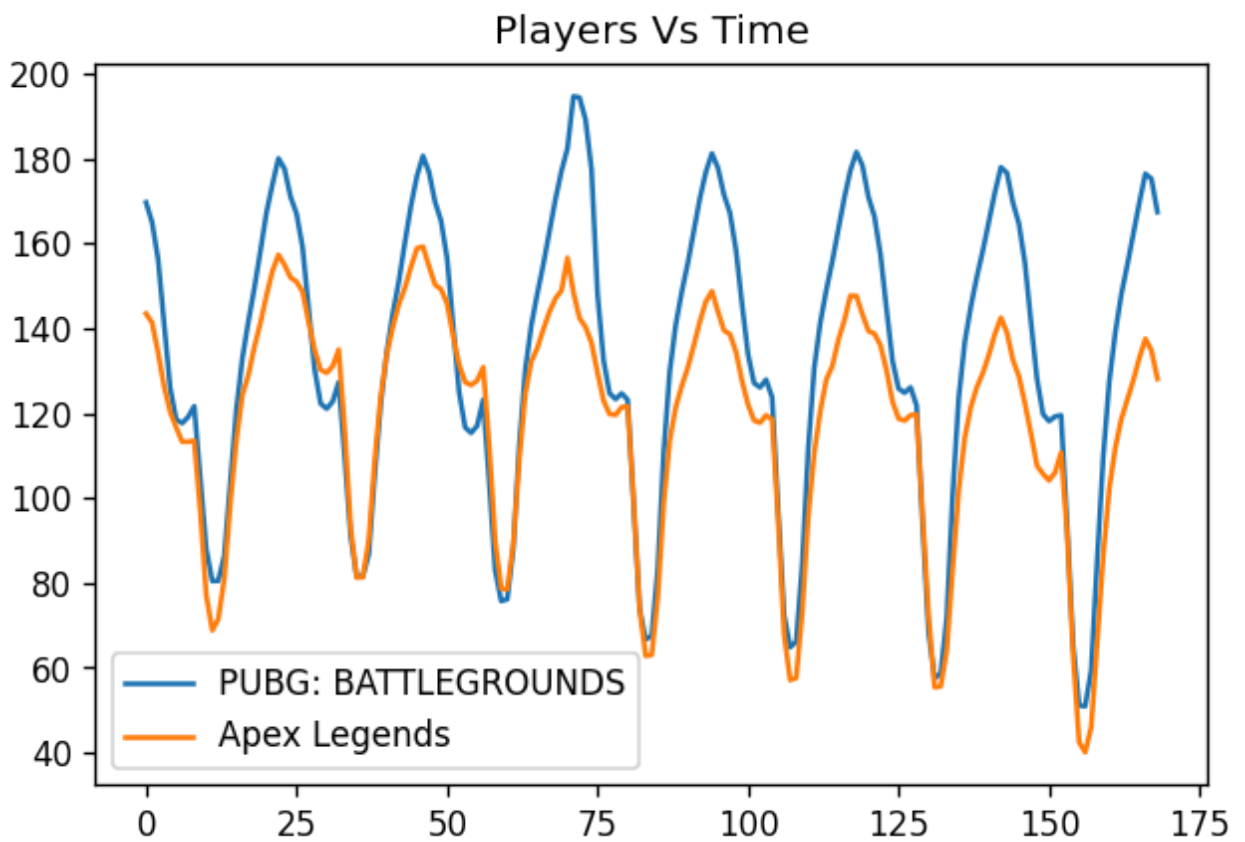
`np.argmax(correlations) = 194`



`np.argmax(correlations) = 162`



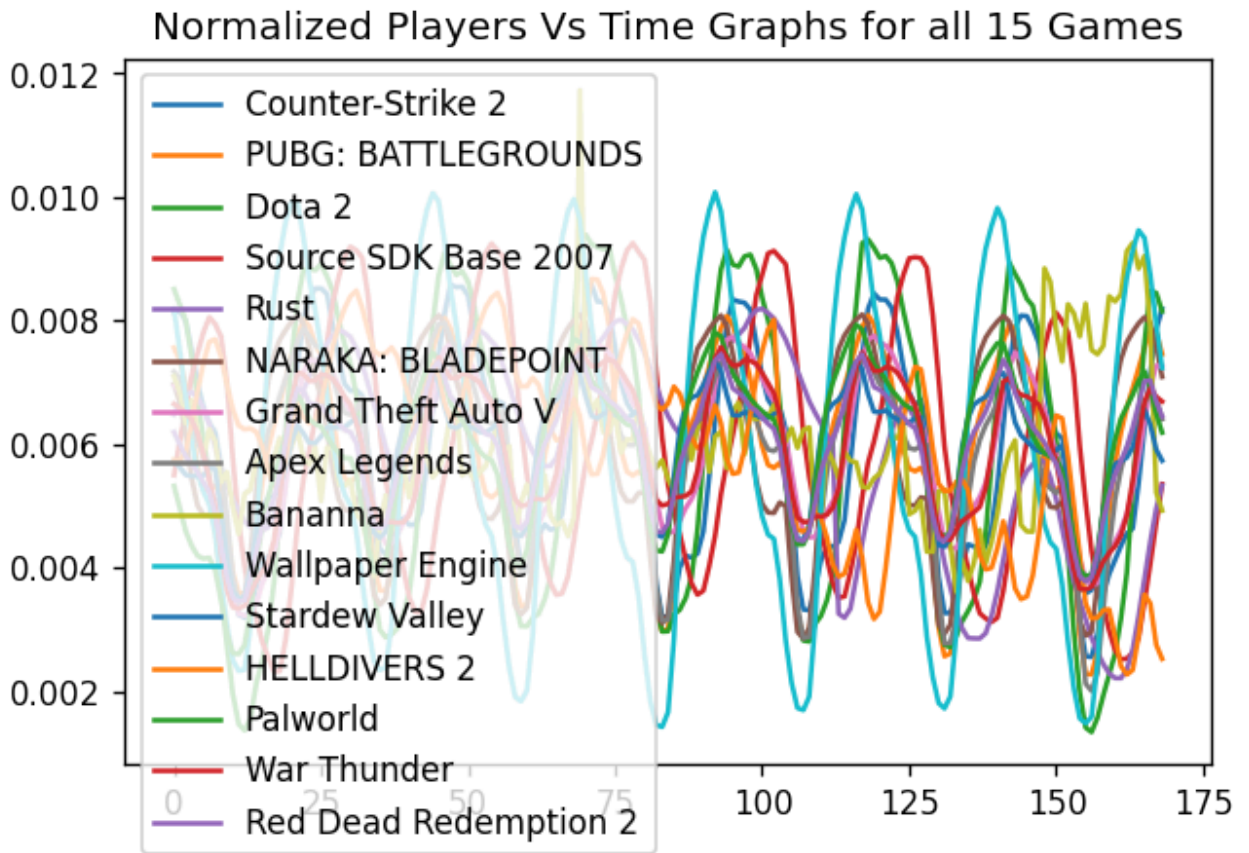
```
np.argmax(correlations) = 22
```



In [170...

```
ordinal = ['1st', '2nd', '3rd', '4th', '5th', '6th', '7th', '8th', '9th', '10th', '11th']
graphs = games[['Players Vs Time'][0]].to_numpy()
names = games['Name'].values.tolist()
```

```
plt.figure(dpi=120)
plt.title(f'Normalized Players Vs Time Graphs for all {len(graphs)} Games')
for i in range(len(graphs)):
    plt.plot(graphs[i]/np.sum(graphs[i]), label=names[i])
plt.legend(loc='upper left')
plt.show()
```



In [215...

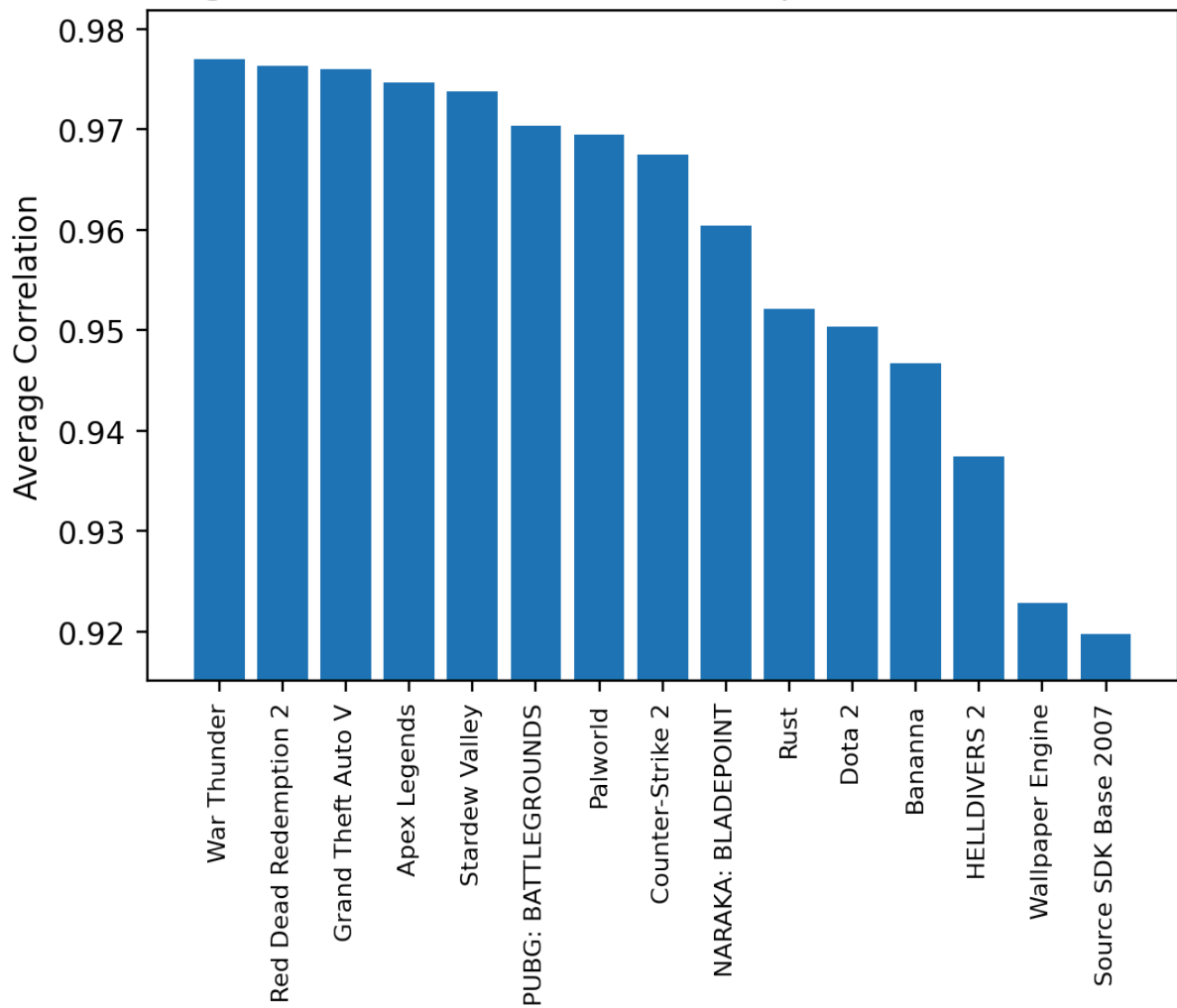
```
def get_correlation(y1, y2):
    r = np.dot(y1, y2) / (np.sqrt(np.sum(y2 * y2)) * np.sqrt(np.sum(y1 * y1)))
    return r

correlations = np.zeros((len(graphs), len(graphs)))
for i in range(len(graphs)):
    for n in range(len(graphs)):
        correlations[i, n] = get_correlation(graphs[i][:N], graphs[n][:N])

avg_correlations = ((np.sum(correlations.copy(), axis=1) - 1) / (len(graphs) - 1)).tolist()
names = games['Name'].values.tolist()
names = sorted(names.copy(), key=lambda n: avg_correlations[names.index(n)], reverse=True)
avg_correlations = sorted(avg_correlations, key=lambda n: n, reverse=True)

plt.figure(dpi=200)
plt.bar(range(len(names)), avg_correlations, bottom=0)
plt.xticks(range(len(names)), labels=names, rotation=90, fontsize=8)
plt.title('Average Correlations Between the Top 50 Games of 2/8/2025')
plt.ylabel('Average Correlation')
plt.ylim(0.995 * min(avg_correlations), 1.005 * max(avg_correlations))
plt.show()
```


Average Correlations Between the Top 50 Games of 2/8/2025



In [213...

```
i = 13
names = games['Name'].values.tolist()
sorted_names = sorted(names.copy(), key= lambda n: correlations[i,names.index(n)], reverse=True)
sorted_correlations = correlations[:,correlations[:, i].argsort()[::-1]]
sorted_correlations = sorted_correlations[correlations[i].argsort()[::-1],:]
sorted_correlations = np.clip(sorted_correlations, 0.95, 1.0)

plt.figure(dpi=200)
plt.imshow(sorted_correlations, cmap='gray')
plt.colorbar()
plt.title(f'Correlations Sorted By Correlation with {names[i]}')
plt.xticks(range(len(sorted_names)), labels=sorted_names, rotation=90, fontsize=6)
plt.yticks(range(len(sorted_names)), labels=sorted_names, fontsize=6)
plt.show()
```

Correlations Sorted By Correlation with War Thunder

